PC HARDWARE ASSIGNMENT

Understanding the parts that make a computer (and your favourite games) run

What is inside this file?

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1 Labeled diagram of a desktop PC

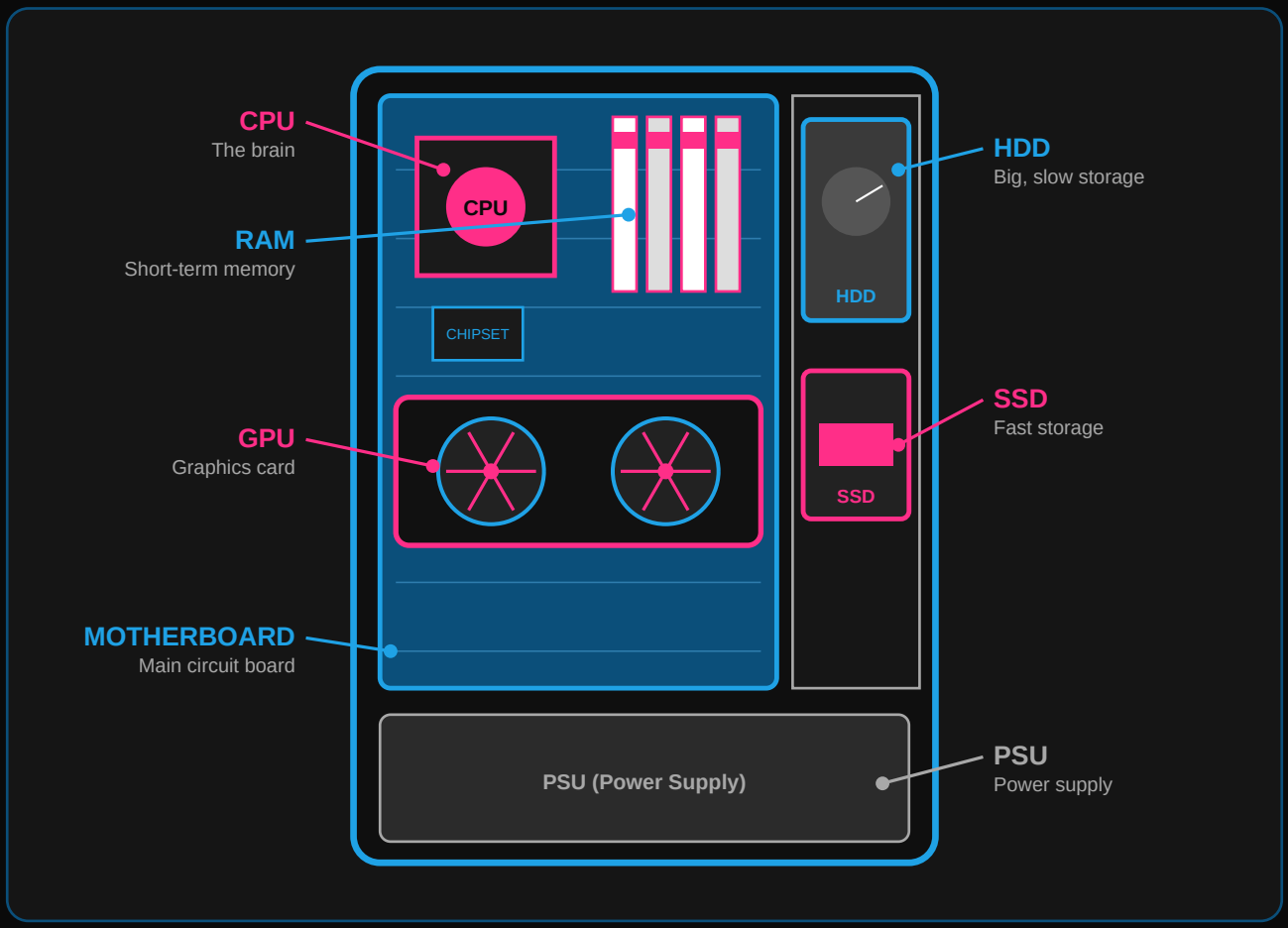

Figure 1: Simple view of the inside of a desktop PC (not to scale).

What does each part do?

Part	Easy explanation
Motherboard	The biggest circuit board in the PC. Every other part plugs into it, so it works like the "road network" that connects everything and lets the parts talk to each other.
CPU	Central Processing Unit. The "brain" of the computer. It follows instructions and does the calculations needed to run programs.
RAM	Random Access Memory. A fast, temporary workspace. It holds the apps and games you are using right now. Its data is erased when the PC is turned off.
GPU	Graphics Processing Unit (graphics card). It creates the pictures, colours and effects you see on the screen. It is very important for gaming and video editing.
SSD	Solid State Drive. Stores files permanently using memory chips with no moving parts. Very fast, so Windows and games load quickly.
HDD	Hard Disk Drive. Stores files permanently on spinning magnetic disks. Slower, but cheap and gives a lot of space for movies, photos and backups.
PSU	Power Supply Unit. Takes electricity from the wall socket and gives the right power to every part.

> **Tip:** In many PCs the SSD can also be a small stick called an **M.2 SSD** that plugs straight into the motherboard. It does the same job as the SSD shown in the drive bay.

2 Comparison table: HDD vs SSD

Both HDD and SSD are used to **store data permanently**, but they work in very different ways. An HDD uses spinning disks and a moving arm (like a tiny record player), while an SSD uses flash memory chips with nothing moving inside.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
How it works	Spinning magnetic disks and a moving read/write head.	Flash memory chips. No moving parts.
Speed	**Slower.** Typical SATA HDDs read roughly 100 to 200 MB/s. Windows can take a minute or more to start.	**Much faster.** SATA SSDs reach about 500 MB/s and NVMe SSDs several thousand MB/s. Windows starts in seconds.
Durability	**Less shock-resistant.** A drop or hard knock can damage the spinning disk or head.	**More shock-resistant.** No moving parts, so it handles bumps and vibration much better.
Noise and heat	Makes spinning and clicking sounds. Uses more power.	Completely silent. Uses less power and stays cooler.
Price per GB	Cheaper. Good for big storage.	More expensive for the same size.
Capacity	Commonly 1 TB to 20 TB or more.	Commonly 256 GB to 2 TB (larger ones cost a lot).
Typical use	Desktop PCs for mass storage: movies, photos, backups, CCTV and servers. Older and budget laptops.	Modern laptops, gaming PCs (OS and games), ultrabooks, phones and tablets (similar flash storage).

Answer to the hint: The **SSD is faster** and the **SSD is more shock-resistant**. SSDs are usually found in laptops and gaming PCs, while HDDs are mostly used in desktops where large cheap storage is needed. Many gamers use **both**: an SSD for Windows and games, and an HDD for movies and backups.

3 Role of CPU, RAM and GPU in games like PUBG or Free Fire

When you play PUBG or Free Fire (using an emulator like Gameloop on a PC), these three parts work together like a team.

CPU

The CPU is the game's "manager". It runs the game's logic: tracking where every player is, calculating bullet hits, handling your keyboard and mouse input, and controlling the actions of other players and the game world. A weak CPU causes lag and stuttering, especially in busy fights with many players.

RAM

RAM is the "short-term memory" where the game keeps the map, characters and sounds it needs right now, so they can be reached instantly. With enough RAM the game runs smoothly and loads maps faster. With too little RAM, the PC slows down and the game may freeze or crash, because it has to keep fetching data from slower storage.

GPU

The GPU is the "artist". It draws every frame you see: the buildings, guns, shadows, smoke and water, many times each second. A better GPU gives higher FPS (frames per second), sharper graphics and smoother gameplay, which helps you spot and aim at enemies more easily.

In one line: The **CPU** decides *what happens* in the game, **RAM** keeps the needed data ready, and the **GPU** shows it all on your screen.

4 Upgrading a PC to run the latest FIFA smoothly

Note: From 2023 the FIFA series by EA was renamed **EA SPORTS FC**, and it is the latest version of that game. The upgrade ideas below apply to both names. Check the official system requirements of the exact game version you want before buying anything.

Components to consider, in order of importance:

1. GPU (Graphics Card) - most important

FIFA has detailed players, stadiums, crowds and lighting. A stronger graphics card gives smoother FPS and lets you use High or Ultra settings at 1080p or even 1440p. If your PC has only integrated graphics or an old card, this gives the biggest boost.

2. CPU (Processor)

The CPU controls player AI, ball physics and the match logic. A modern 6-core processor avoids stuttering and keeps the game steady, and stops the CPU from "holding back" a new graphics card (called a bottleneck).

3. RAM

8 GB is the bare minimum today. Upgrading to **16 GB** (dual channel, i.e. two sticks) makes the game and Windows run together without slowing down. It is a cheap and easy upgrade.

4. Storage: switch to an SSD

Installing the game on an SSD makes loading, menus and match start-up much faster than on an HDD. It is also a very noticeable upgrade for the whole PC.

5. PSU (Power Supply)

A new GPU needs more power. Check that your PSU has enough wattage and the right connectors. A good, reliable PSU also protects all other parts.

6. Cooling and Monitor (optional)

Better cooling stops the CPU and GPU from getting too hot and slowing down. A 144 Hz monitor helps you see the smooth FPS you are paying for.

Conclusion: If the budget is small, first upgrade to an **SSD** and **16 GB RAM**, then the **GPU**. Then check that the **CPU** and **PSU** can support it. Always check the game's official requirements first.